

CO-PO-PSO Mapping & Attainment

CS301 — Data Structures

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Review required before filing. *This document is a faculty-review **draft**, not an authoritative NBA/AICTE filing. The CO-PO-PSO correlation matrix (Section 3) reflects each Course Outcome’s stated content, not verified pedagogical practice, and every numeric attainment value (Section 4) is marked [FILL] — no student assessment score data exists in this repository. The instructor must review, correct, and sign off on both before submission. Only Week 1 of this course’s lecture content has been built at the time this draft was generated (CLAUDE.md project-state table) — the CO statements below are drawn from the syllabus, not from delivered content, for the weeks not yet built.*

1 Course Outcomes

Numbered here for the first time from `syllabi/CS301.md`’s Learning Objectives, in delivery order. Bloom’s-taxonomy levels are a first-pass draft inferred from each objective’s stated verb; objectives whose verbs span two levels are flagged rather than silently collapsed to one.

2 Assessment-Tool Mapping

Cross-referenced against `syllabi/CS301.md`’s own Weekly schedule and Assessment tables (Class Tests 1–3, Assignments 1–4, Practical/lab P1–P13, End-semester exam) — no new assessment data is invented here, only cross-referenced.

CO	Statement	Bloom level	Weeks
CO1	Manage dynamic memory in C with pointers (<code>malloc/calloc/realloc/free</code>) and express a data structure as an abstract data type separated from its implementation	Understand/Apply (flag: “express”)	1
CO2	Derive the space and time complexity of an algorithm, express it in the appropriate asymptotic notation, and distinguish best-, worst-, and average-case behaviour	Analyze	2
CO3	Implement stacks and queues (linear, circular, double-ended, priority) over arrays, and apply them to expression conversion, expression evaluation, and recursion problems	Apply	3–5
CO4	Implement singly, doubly, and circular linked lists, and select between array and linked representations for a given access pattern	Apply/Evaluate (flag: “select”)	6–7
CO5	Apply and analyze linear and binary search, and implement the elementary sorting algorithms with their cost analysis	Analyze	8
CO6	Implement and compare the efficient sorting algorithms (quick, merge, shell, radix, heap), and justify a choice against input characteristics	Evaluate (flag: “compare”, “justify”)	9
CO7	Select a file organization and design a hash function with an appropriate collision-resolution scheme	Create (flag: “design”)	10
CO8	Implement binary trees, binary search trees, and AVL trees with their traversals and rebalancing rotations	Apply	11
CO9	Represent graphs, traverse them with BFS and DFS, and apply Prim’s, Kruskal’s, and Dijkstra’s algorithms to spanning-tree and shortest-path problems	Apply	12

Table 1: Course Outcomes, copied verbatim from `syllabi/CS301.md`.

CO	Assessed in
CO1	Assignment 1, End-semester exam
CO2	Class Test 1 (Unit I), End-semester exam
CO3	Assignment 2, Class Test 1 (Unit I), End-semester exam, Labs P1–P6
CO4	Assignment 2, Class Test 2 (Unit II), End-semester exam, Lab P7
CO5	Assignment 3, Class Test 2 (Unit II), End-semester exam
CO6	Assignment 3, End-semester exam, Labs P9–P11
CO7	Assignment 3, End-semester exam
CO8	Assignment 4, Class Test 3 (Unit III), End-semester exam, Labs P8, P12
CO9	Assignment 4, End-semester exam, Lab P13

Table 2: Every CO maps to at least one existing assessment tool – no unassessed CO.

3 CO-PO-PSO Correlation Matrix

Review required before filing. *Draft correlation matrix — requires instructor review and sign-off before submission. Correlation strengths (1 = slight, 2 = moderate, 3 = substantial, blank = no correlation) reflect the CO's stated content, not verified pedagogical practice. PSO columns are [FILL]: Program-Specific Outcomes are department-defined and not yet standardized anywhere in this repository.*

NBA mandates exactly these 12 Program Outcomes for every accredited UG engineering program: PO1 Engineering knowledge, PO2 Problem analysis, PO3 Design/development of solutions, PO4 Conduct investigations of complex problems, PO5 Modern tool usage, PO6 The engineer and society, PO7 Environment and sustainability, PO8 Ethics, PO9 Individual and team work, PO10 Communication, PO11 Project management and finance, PO12 Life-long learning.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2										1	[FILL]	[FILL]
CO2	3	3										1	[FILL]	[FILL]
CO3	3	2	2										[FILL]	[FILL]
CO4	2	2	2										[FILL]	[FILL]
CO5	3	3											[FILL]	[FILL]
CO6	2	3	1										[FILL]	[FILL]
CO7	2	2	3										[FILL]	[FILL]
CO8	3	2	2										[FILL]	[FILL]
CO9	2	3	2		1								[FILL]	[FILL]

Table 3: Draft CO-PO-PSO correlation matrix (1/2/3 scale; blank = no correlation).

4 Attainment Methodology

Standard NBA formulas, with every numeric slot left as [FILL] — no student assessment score data exists in this repository. Once real per-student, per-question scores exist, mapped to the CO each question tests via Table 2, the direct-attainment percentages below can be computed and substituted.

- **Direct attainment (per CO):** percentage of students scoring at or above the target threshold (typically 60%) in the assessment(s) mapped to that CO. Attainment level: Level 3 if [FILL]% of students meet the target, Level 2 if [FILL]%, Level 1 if [FILL]% (institute-specific cutoffs, not yet fixed in this repository).
- **Indirect attainment (per CO):** from a course-exit survey (student self-assessment), normalized to the same 1–3 scale. [FILL] — no survey instrument or data exists yet.
- **Overall CO attainment:** $0.8 \times \text{direct} + 0.2 \times \text{indirect}$ (weights are institute policy; adjust if your department's weighting differs).
- **PO attainment:** for each PO, the weighted average of the overall CO attainments mapped to it, using the correlation levels in Table 3 as weights:

$$\text{PO}_j \text{ attainment} = \frac{\sum_i \text{corr}(\text{CO}_i, \text{PO}_j) \times \text{overall attainment}(\text{CO}_i)}{\sum_i \text{corr}(\text{CO}_i, \text{PO}_j)}$$

5 Faculty Sign-Off

I have reviewed the Course Outcomes (Section 1), the assessment-tool mapping (Section 2), and the CO-PO-PSO correlation matrix (Section 3) above, and confirm they accurately reflect this offering of CS301.

Name: _____

Date: _____